



Reading: Exemplar: Qwiklabs assessment: Managing services in Windows

Introduction

As a system administrator, you'll need to know how to look at the status of a running service, and how to stop, start, and restart running services. You'll also need to know how to configure services and fix problems you may encounter while running them.

In this lab, you looked at the list of services that were running on a Windows machine. You practiced stopping and starting some of these services, as well as querying their status. You also enabled serving web pages and customizing the configuration of the web server.

This exemplar is a walkthrough of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.

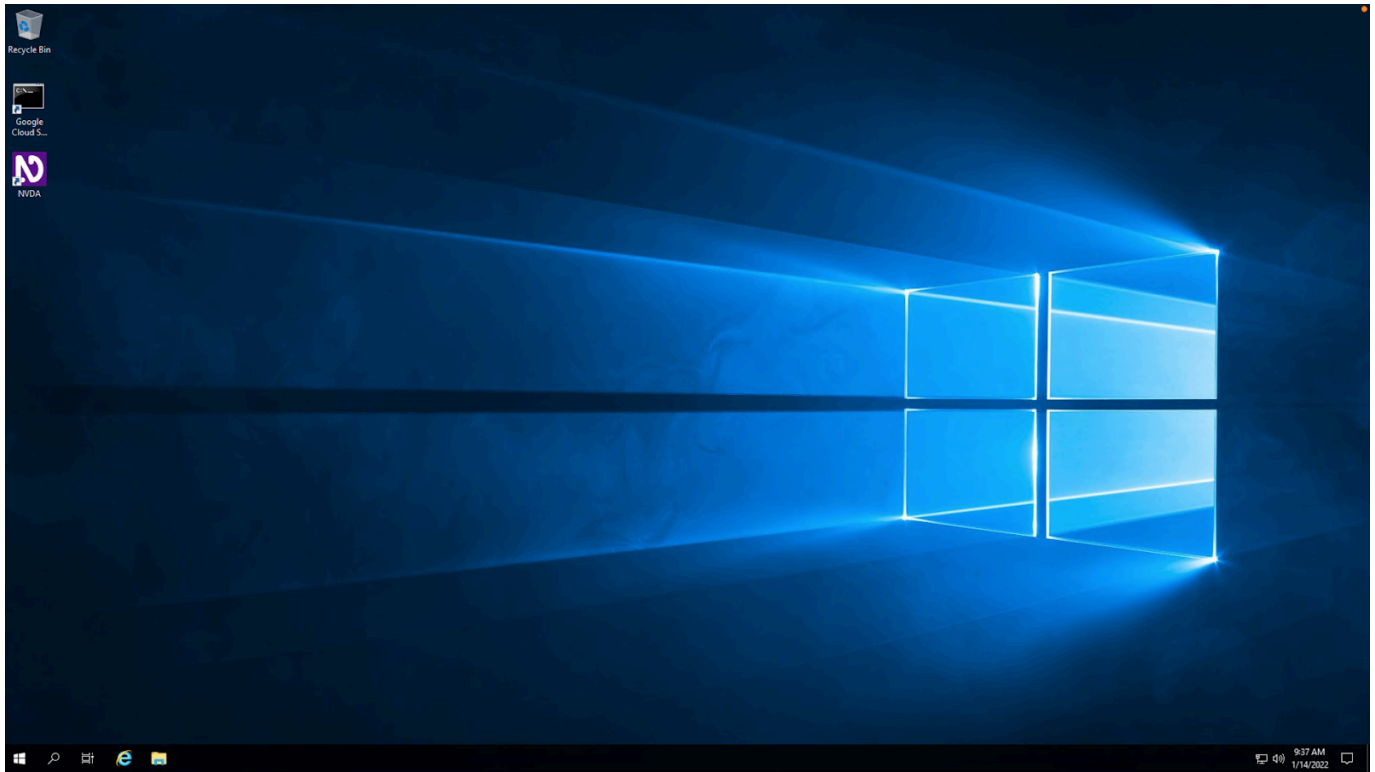
A green rectangular button with rounded corners and the text "Start Lab" in white.

After you click the **Start Lab** button, you will see a panel appear below where the start lab button was that has an **Open Windows VM** button.

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

A light blue rectangular button with rounded corners and a thin blue border, containing the text "Open Windows VM" in blue.

Click the **Open Windows VM** button and a new tab will open with a visual interface for Windows OS, where you will be performing further steps in the lab. You should have a visual interface for Windows that looks like this:



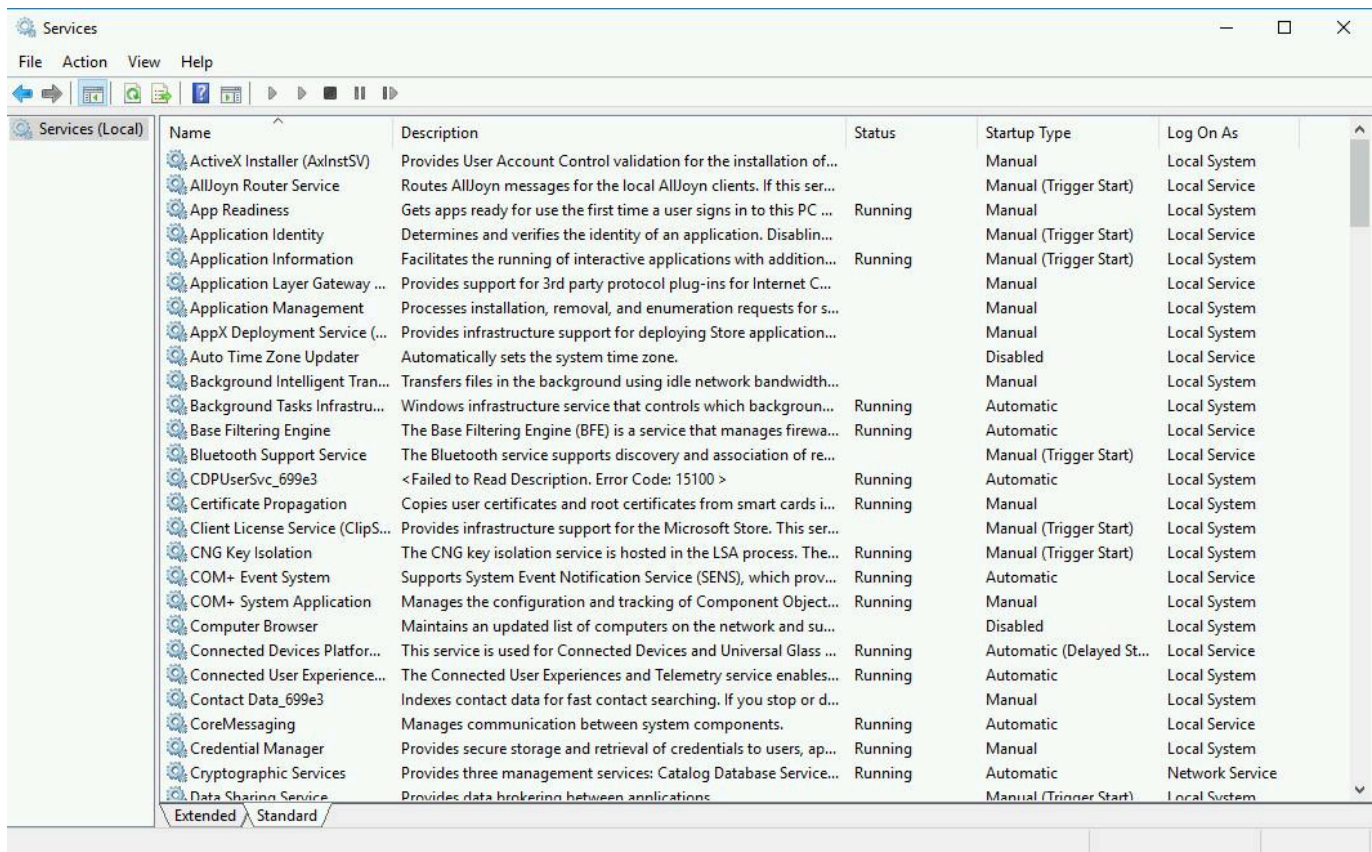
Learning tip:

During this lab, we will show how to do certain tasks using graphical tools as well as using the command line with PowerShell. While you can copy and paste the commands that are presented throughout the lab, we recommend typing them out manually, to help with understanding and remembering these commands.

List system services

First, we will connect to the **Services** application that is provided by Windows as an administrative tool. In order to open this application, you can either open the **Control Panel**, click on **System and Security**, then on **Administrative Tools** and finally on **Services**; or you can type "Services" into the Windows menu.

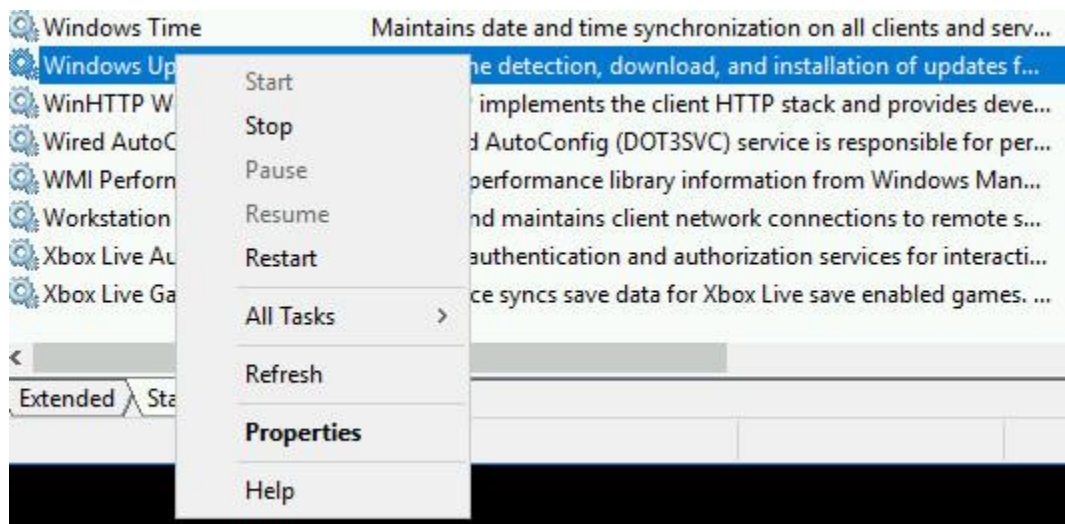
Once you open the application, make sure you make the columns wider so that you can see the actual descriptions of the services that are available in the system. You can click on one of the columns to change the order and, for example, see all services that have **Running** as their **Status** together.



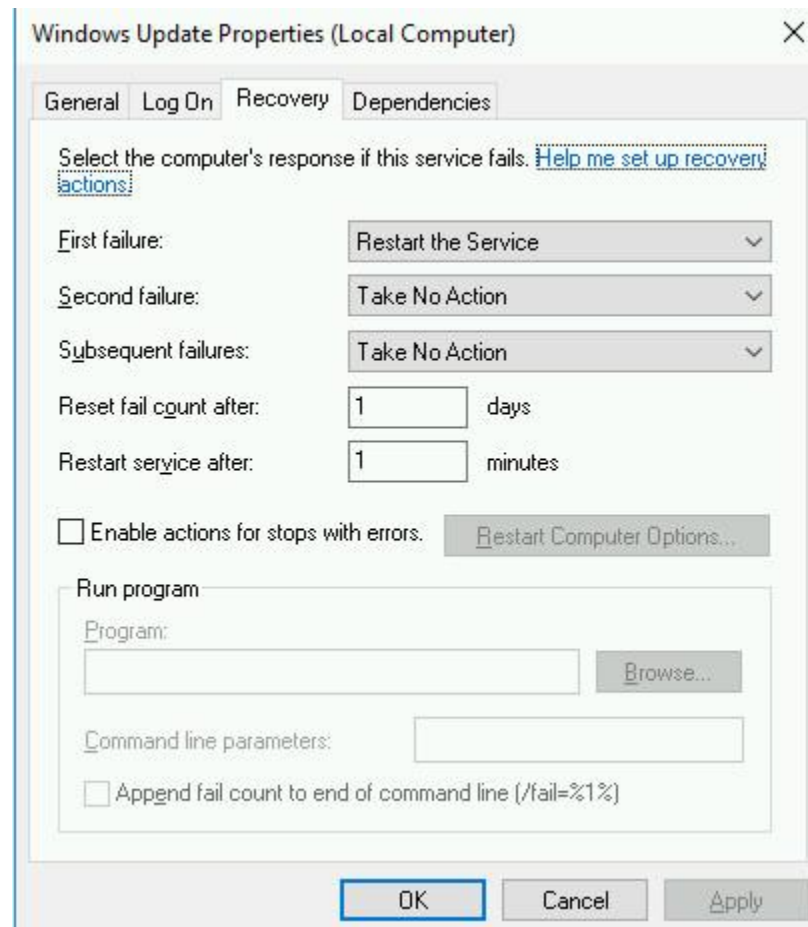
From this application, you can start and stop services that are running in the system.

Caution: keep in mind that some of the services that are running may be critical to ensure that you stay connected to the machine. Pay attention to what the services do before you try stopping them.

When you right-click on the line that shows a service, you get a menu of action options. The menu shows all actions, but the ones that are not currently available are grayed out. So, if the service is running, **Start** will be grayed out. If the service is not running, **Stop** will be grayed out.



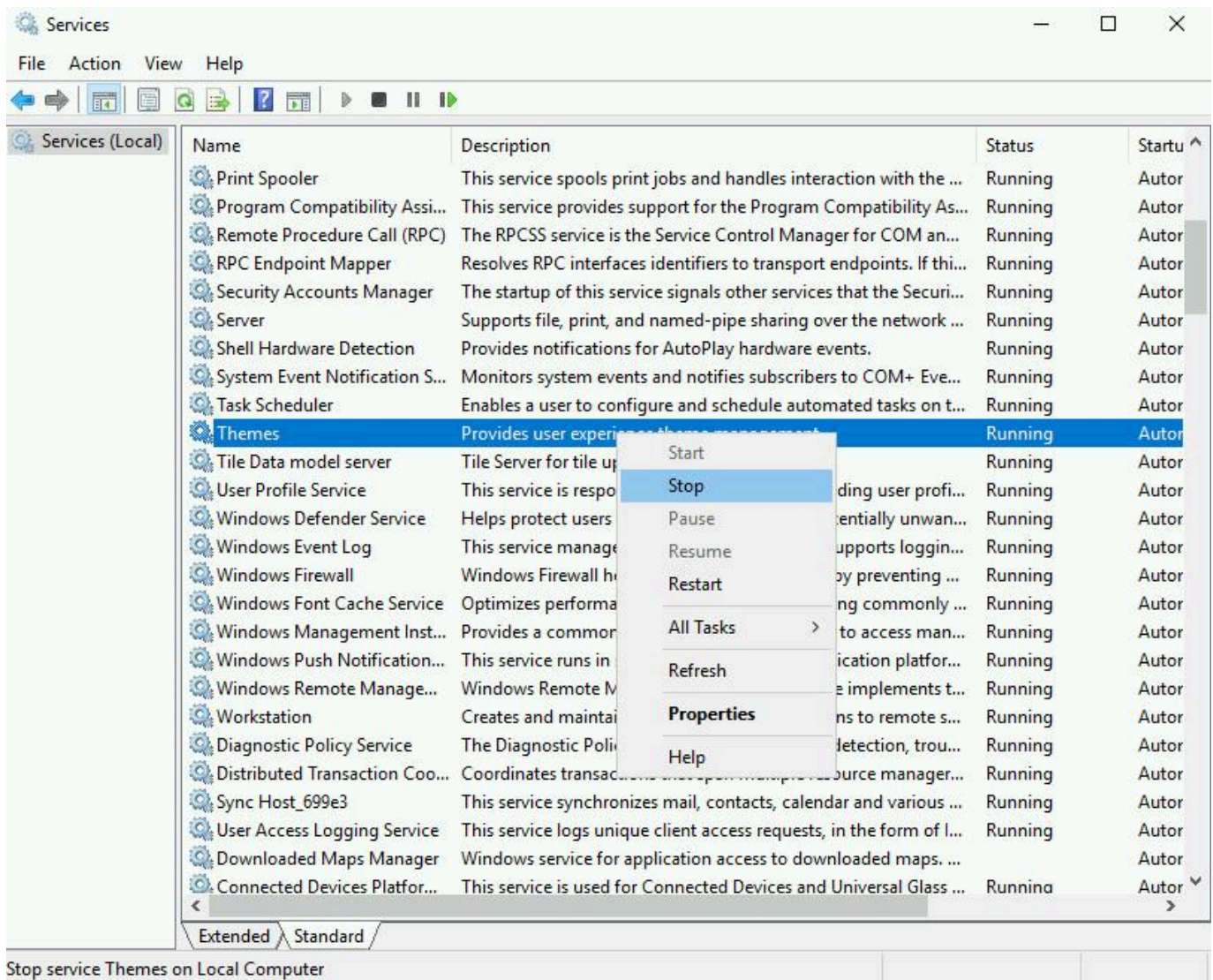
You can also access the **Properties** menu, where you will see other data and configuration about the service. For example, in the **Recovery** tab, it will indicate what's the configured behavior in case of a service failing.



Stopping and starting services

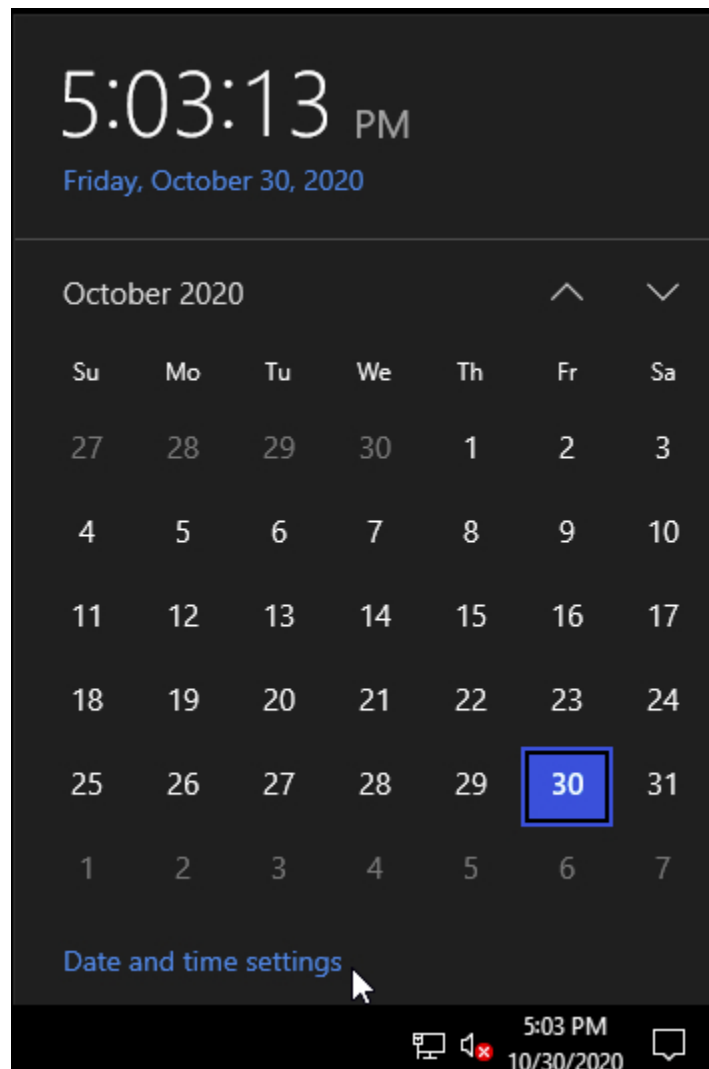
Let's look at the **Themes** service. This service is in charge of managing the graphical themes applied to the machine. It's also in charge of managing certain features related to the video card. While you wouldn't want to stop it on your local physical machine as it might cause problems with your graphical interface, it's fine to stop it for the purposes of this lab as it's a virtual machine with a virtualized video card.

So, go ahead and right-click on the **Themes** line and then click **Stop** in the pop-up menu. A window will appear showing the progress of stopping the service and once it's gone, the service will be stopped.

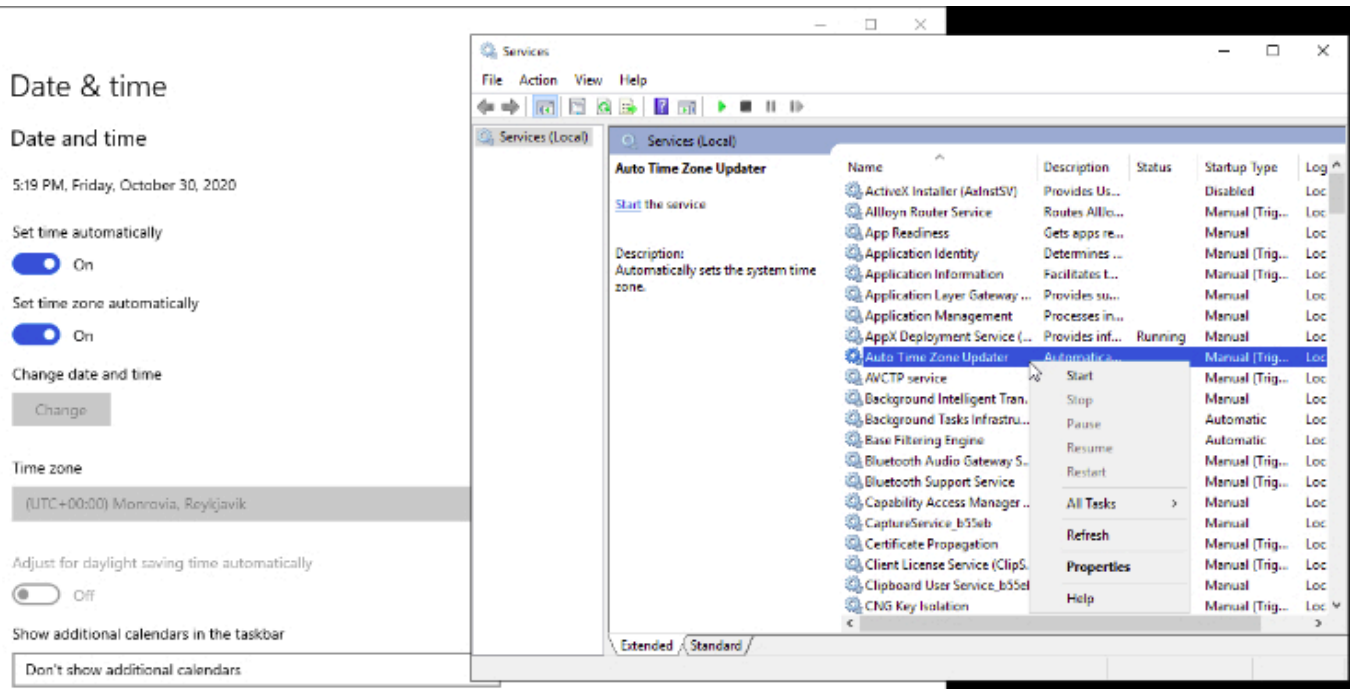


Let's now look at the **Performance Logs & Alerts** service. This service is not running by default. It can be used to collect logs related to the performance of the machine and to setup alerts that will trigger when certain thresholds are passed. The data collected by this service can be configured and visualized in the **Server Manager** application.

Let's now start this service: right-click on the **Performance Logs & Alerts** line and then click **Start** in the pop-up menu. A similar window as before will appear, showing the progress and once it's gone, the service will be started.



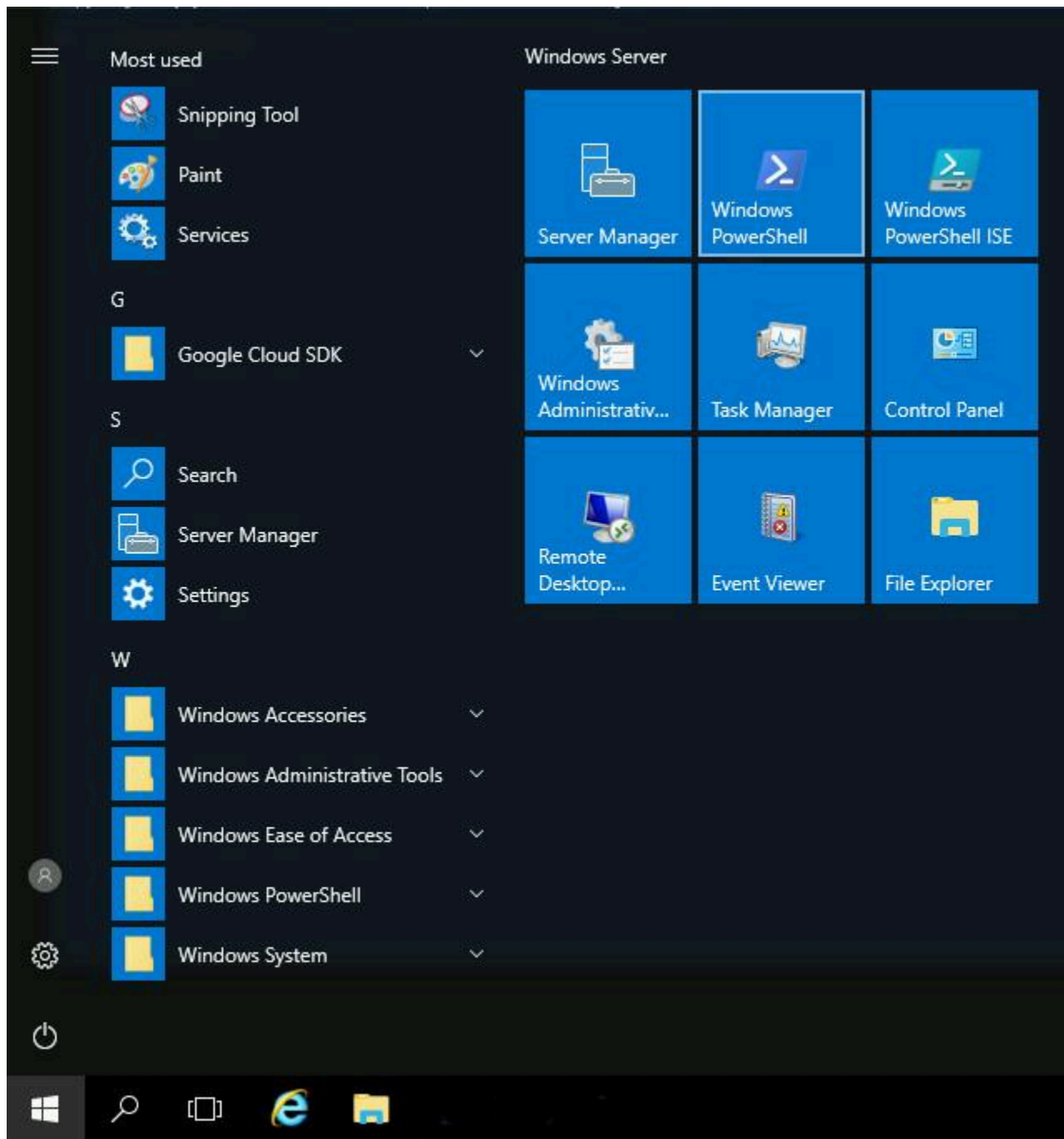
If you click on that link, the settings window for all configuration related to the date and time will appear, including the **Set time zone automatically** setting. This setting is directly related to the **Auto Time Zone Updater service**. Click on the slider to enable the setting, go back to the **Services** application, click **Refresh** in the right-click menu, and you'll see that the service is now enabled.



Interacting with services through PowerShell

While as a user it's common to use Graphical User Interfaces (or GUIs) to interact with the system, as a system administrator you will usually want to automate the tasks that you do, and for that you will be using the command line.

Let's see how to perform these same tasks through the command line. Go ahead and open the **Windows PowerShell** application, by clicking on the **Start** menu and then selecting it from the list of applications.



In this shell, we can use the **Get-Service** command to list all the available services.

Get-Service

OUTPUT:

Status	Name	DisplayName
-----	----	-----
Stopped	VSS	Volume Shadow Copy
Running	w32time	Windows Time
Stopped	WaaSMedicSvc	Windows Update Medic Service
Stopped	WalletService	WalletService
Stopped	WarpJITSvc	WarpJITSvc
Stopped	WbioSrv	Windows Biometric Service
Running	WcmSvc	Windows Connection Manager
Stopped	WdiServiceHost	Diagnostic Service Host
Stopped	WdiSystemHost	Diagnostic System Host
Running	WdNisSvc	Windows Defender Antivirus Network ...
Stopped	WeSvc	Windows Event Collector
Stopped	WEPHOSTSVC	Windows Encryption Provider Host Se...
Stopped	wercplsupport	Problem Reports and Solutions Contr...
Stopped	WerSvc	Windows Error Reporting Service
Stopped	WiaRpc	Still Image Acquisition Events
Running	WinDefend	Windows Defender Antivirus Service
Running	WinHttpAutoProx...	WinHTTP Web Proxy Auto-Discovery Se...
Running	Winmgmt	Windows Management Instrumentation
Running	WinRM	Windows Remote Management (WS-Manag...
Stopped	wisvc	Windows Insider Service
Stopped	wlidsvc	Microsoft Account Sign-in Assistant
Stopped	wmiApSrv	WMI Performance Adapter
Stopped	WMPNetworkSvc	Windows Media Player Network Sharin...
Stopped	WPDBusEnum	Portable Device Enumerator Service
Running	WpnService	Windows Push Notifications System S...
Running	WpnUserService_...	Windows Push Notifications User Ser...
Stopped	WSearch	Windows Search
Stopped	wuauserv	Windows Update

This is a long list. You can scroll up using **PgUp/PgDown**, or with the mouse wheel. This list includes the **Status** of the service, the **Name** and the **DisplayName**. The **Name** is short and is the one used to get information of a specific service, the **DisplayName** is a more descriptive and longer name.

For example, we can get information of the **wisvc** service:

```
Get-Service wisvc
```

OUTPUT:

Status	Name	DisplayName
-----	----	-----
Stopped	wisvc	Windows Insider Service

This tells us that **wisvc** is the short name of the **Windows Insider Service**. This service allows you to test out the latest OS builds and Apps from Microsoft. We can get more information for the service, using the **Format-List** command.

```
Get-Service wisvc | Format-List *
```

OUTPUT:

```
Name                : wisvc
RequiredServices    : {rpcss}
CanPauseAndContinue : False
CanShutdown         : False
CanStop             : False
DisplayName          : Windows Insider Service
DependentServices   : {}
MachineName         : .
ServiceName         : wisvc
ServicesDependedOn  : {rpcss}
ServiceHandle       : SafeServiceHandle
Status              : Stopped
ServiceType         : Win32ShareProcess
StartType           : Manual
Site                :
Container           :
```

Stopping and starting services

Now, we can start the **Windows Insider Service**, using the **Start-Service** command:

```
Start-Service wisvc
```

This command doesn't give any output. To verify that it worked, let's get the status again, using **Get-Service**:

```
Get-Service wisvc
```

OUTPUT:

Status	Name	DisplayName
-----	----	-----
Running	wisvc	Windows Insider Service

The service is now started. But this is an important service, we don't want it to stay started; let's stop it using the **Stop-Service** command.

```
Stop-Service wisvc
```

Again, this doesn't give any output if it succeeds, let's use **Get-Service** to see if it worked.

```
Get-Service wisvc
```

Yes, it's stopped again.

Enabling services

We can stop and start any services that are already enabled in the system, but not services that are disabled. Let's look at one example of a service that is currently disabled. We will use the **ScardSvr** or **Smart Card** service. This service is used to read smart cards and it's currently disabled because it's not needed. We can see that it's disabled, by using the **Get-Service** plus **Format-List** commands:

```
Get-Service ScardSvr | Format-List *
```

OUTPUT:

```
Name           : ScardSvr
RequiredServices : {}
CanPauseAndContinue : False
CanShutdown    : False
CanStop        : False
DisplayName     : Smart Card
DependentServices : {}
MachineName     : .
ServiceName     : ScardSvr
ServicesDependedOn : {}
ServiceHandle   :
Status          : Stopped
ServiceType     : Win32ShareProcess
StartType       : Disabled
Site           :
Container       :
```

The **StartType** field says that this service is **Disabled**. This means that it can't be started until it's enabled. Let's see what happens if we try to start it anyway:

```
Start-Service ScardSvr
```

OUTPUT:

```
Start-Service : Service 'Smart Card (ScardSvr)' cannot be started due to the
following error: Cannot start service
ScardSvr on computer '.'.
At line:1 char:1
+ Start-Service ScardSvr
+ ~~~~~
+ CategoryInfo          : OpenError:
(System.ServiceProcess.ServiceController:ServiceController) [Start-Service],
ServiceCommandException
+ FullyQualifiedErrorId :
CouldNotStartService,Microsoft.PowerShell.Commands.StartServiceCommand
```

Instead of the usual empty output, we get an error message. The error says that the service couldn't be started. To change that, we will use the **Set-Service** command, which allows us to modify some of the properties of the service.

Set-Service ScardSvr -StartupType Manual

This won't generate any output, but if we now try to start the service, it will succeed.

Start-Service ScardSvr

This time we got no output, which means that there were no errors. Let's check the status one last time:

Get-Service ScardSvr

OUTPUT:

Status	Name	DisplayName
Running	ScardSvr	Smart Card

Well done! We have enabled and started the service.

Enabling additional features

Enabling additional features

There are a number of features that are available in a Windows system that are not enabled by default. This is so that the system administrator only has to manage the features that are actually in use, instead of all the possible features out there.

We can use the **Install-WindowsFeature** command to enable these features. For example, let's run this command to enable the few more web serving features in the system:

```
Install-WindowsFeature Web-WebServer,Web-Mgmt-Tools -IncludeAllSubFeature
```

OUTPUT:

Collecting data...
10%
[000000000000000]

10%

[0000000000000000

1

Be patient, this will take a few minutes to run, as it's downloading additional components and then installing them on the system. You will see a progress bar on the console, although not necessarily at the cursor.

When the progress bar disappears, it should tell you that it succeeded in installing the service which is now running.

OUTPUT:

Success	Restart	Needed	Exit Code	Feature Result
-----	-----	-----	-----	-----
True	No		Success	{.NET Framework 3.5 (includes .NET 2.0 and...

There are a number of new services that were added when enabling this feature. For example, **IISADMIN** is the service used to publish websites on the machine.

```
Get-Service IISADMIN
```

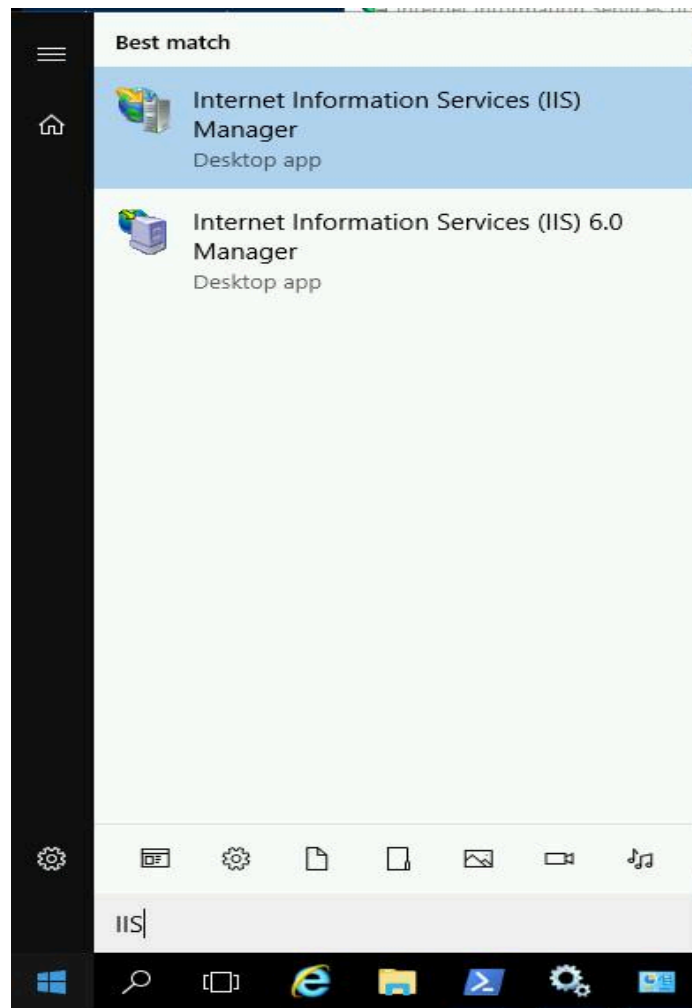
OUTPUT:

Status	Name	DisplayName
-----	----	-----
Running	IISADMIN	IIS Admin Service

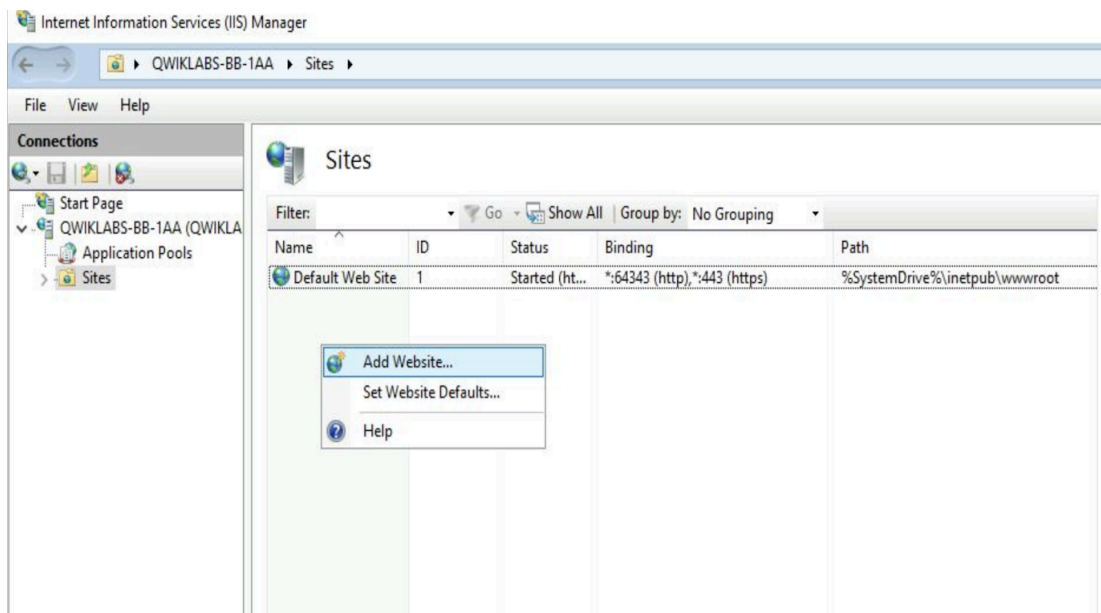
With this service running, you'll be able to use the web serving functionality on the machine.

Serving web pages

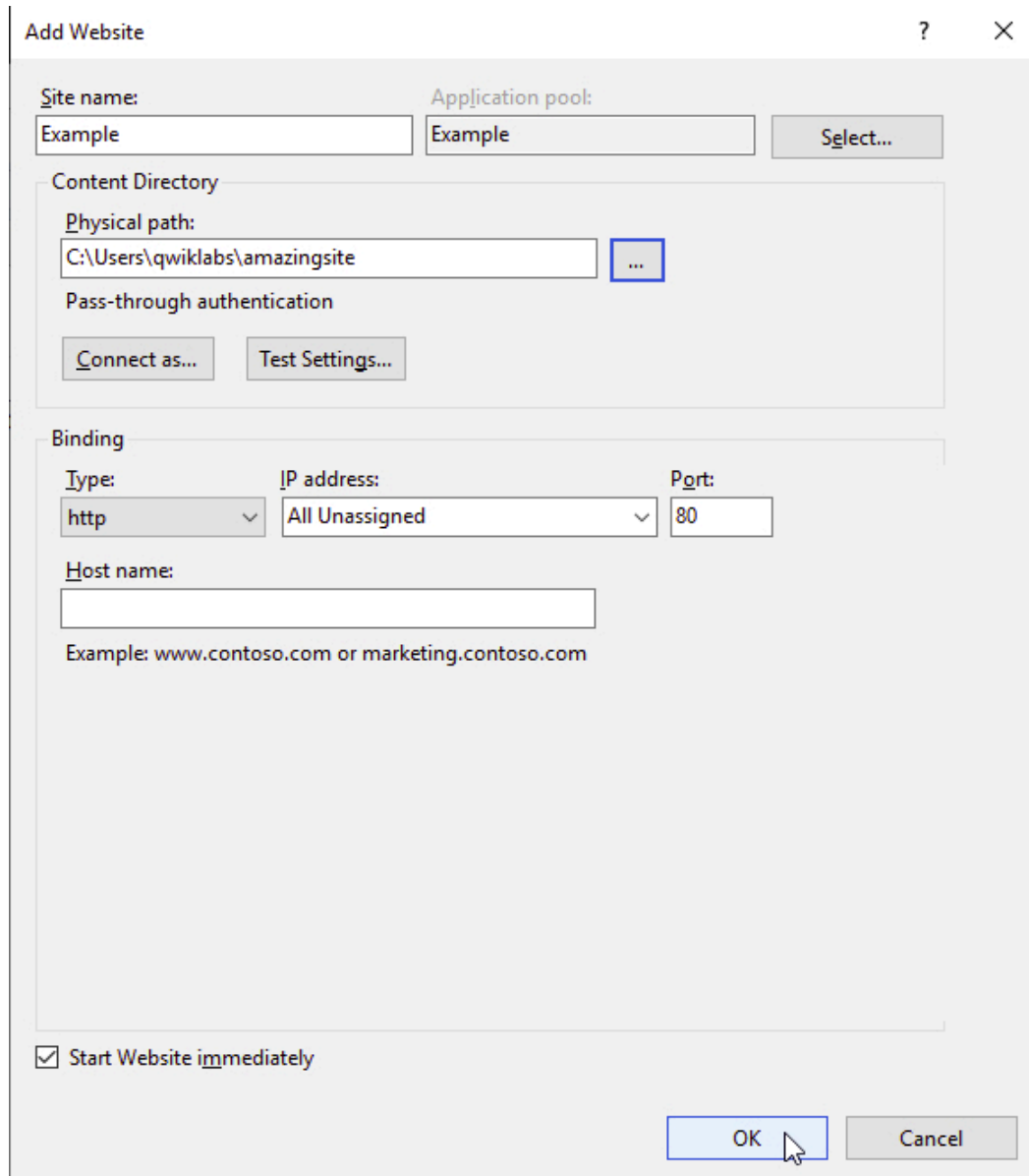
Let's configure our newly installed web server to serve our own website. In order to do this, you'll need to open the **Internet Information Services (IIS) Manager** program. To do that, type **IIS** into the Windows menu.



Once the application opens, expand the **QWIKLABS-BB-XXX** option, and click on the **Sites** option. This shows a list of available sites, currently only showing the **Default Website**.



Let's add a new site, by right-clicking on the list and selecting **Add Website**. This will pop up a window with some fields that need to be filled in. Naming the site enables you to identify it from others. You can call it whatever you want. The physical path of the website is the location where the files are located in the computer. In this case, the files are located in **C:\Users\qwiklabs\amazingsite**, so select that folder as the path.



The screenshot shows the 'Add Website' dialog box with the following fields and controls:

- Site name:** Example
- Application pool:** Example, with a **Select...** button.
- Content Directory:**
 - Physical path:** C:\Users\qwiklabs\amazingsite, with a browse button (...).
 - Pass-through authentication:** (disabled)
 - Connect as...** and **Test Settings...** buttons.
- Binding:**
 - Type:** http (dropdown)
 - IP address:** All Unassigned (dropdown)
 - Port:** 80
 - Host name:** (empty text box)
 - Example: www.contoso.com or marketing.contoso.com
- ☒ **Start Website immediately**
- OK** and **Cancel** buttons at the bottom right.

Click **OK** after filling all the provided details, then you will see that your website was created and running on **port 80**. Now you can verify the new website by pointing your browser to the **External IP address** shown in the **Connection Details Panel**.

Amazing Results

This website shows that it's possible to have amazing results and share them with others.

Woohoo!

Conclusion

Congratulations! You've successfully listed all the services that are available on the machine, practiced stopping and starting some of these services, and checked their status. You also configured the machine to serve web pages by enabling an additional system feature, and then customized this to serve a different website.

These are important skills that are essential to your future work as a system administrator. Keep it up!
